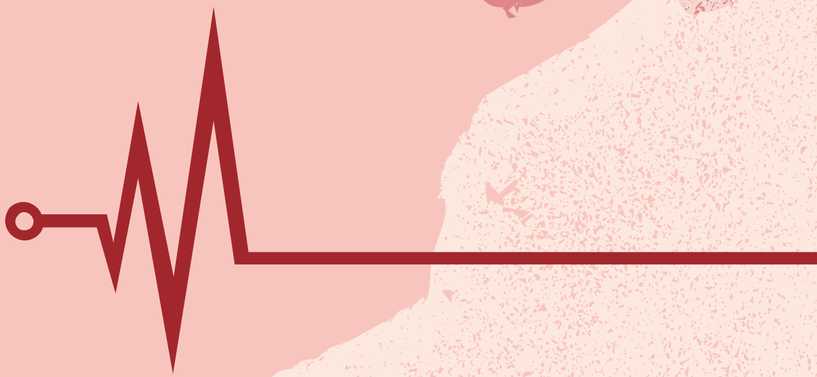
RED CONNECT

TEAM NOOBBYTES

PROBLEM STATEMENT

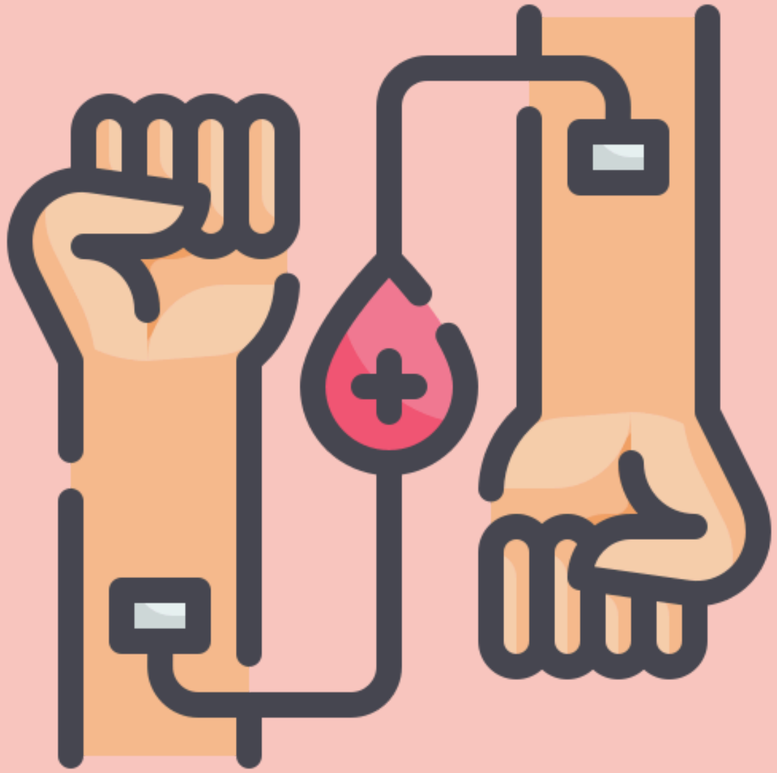
The proposed Blood Donation and Emergency Help app aims to provide a comprehensive and immediate solution to blood shortages in critical situations. By connecting donors with hospitals and individuals in need, the app leverages real-time notifications and intelligent donor matching based on blood type, location, and eligibility. Additionally, it incorporates a robust blood bank inventory management feature that enables hospitals to track blood units, manage availability, and request specific blood types when necessary. The app also integrates machine learning for health eligibility prediction, offers scheduling for donations, and provides first responders with real-time alerts. This holistic approach ensures efficient management of blood donations and inventory, ultimately reducing shortages and saving lives during emergencies.





RED CONNECT

Donate Blood, Save Lives!



WHY THIS IDEA

The idea for the Blood Donation and Emergency Help app is driven by the urgent need to address blood shortages in critical situations. By connecting donors with hospitals and individuals in need, the app facilitates real-time communication and intelligent donor matching based on blood type and location. It enhances inventory management for hospitals, allowing them to track blood units and request specific types efficiently. Additionally, the integration of machine learning for health eligibility prediction streamlines the donation process, while real-time alerts for first responders improve emergency response capabilities. This holistic approach promotes community engagement in blood donation, ultimately saving lives and improving health outcomes.

FEATURES





Real-Time Blood Request Alerts

Immediate notifications for nearby blood donation requests, including blood type, hospital location, and urgency.

Machine Learning Integration for Eligibility Prediction

Use of ML algorithms to assess donor health and eligibility based on various factors.





Scheduled Donation Appointments

Option for donors to schedule appointments at local blood banks or hospitals, with reminders and preparation guidelines

Blood Bank Inventory Management

A dashboard for hospitals to manage blood inventory, track blood units, and request specific blood types.



Geolocation Services

Integrated Google Maps for locating nearby hospitals, blood banks, and donation drives.



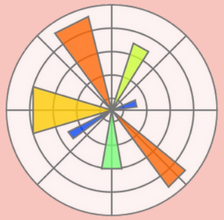
Cloud Hosting & Scalability

Cloud-based backend for scalable hosting and management, ensuring efficiency during emergencies.

TECH STACK



Kotlin and XML



Matplotlib



Firebase



Pandas



Google Maps API



Flask

IMPLEMENTATION DETAILS

Alert Messaging System Implementation

- Blood Request Creation: Hospitals or individuals create a blood request with details such as blood type, urgency, and location. This triggers the system to begin donor search.
- Donor Matching: A backend algorithm identifies nearby eligible donors based on blood type, location (using geolocation services), and availability.
- Real-Time Notifications: Push notifications are sent to matched donors using Firebase Cloud Messaging (FCM) with the necessary request details. SMS notifications through Twilio can serve offline users.
- Response and Appointment Scheduling: Donors can accept the request in the app and schedule a donation appointment at the nearest blood bank or hospital.

IMPLEMENTATION DETAILS

Machine Learning for Eligibility Prediction

Data Preparation:

- Utilized features: months since last donation, total number of donations, total blood donated in c.c., and months since first donation
- Feature engineering: Created 'donation frequency' feature (donations per month)

Data Preprocessing:

- Applied Standard Scaler to normalize feature scales

Model Selection and Training:

- Compared Logistic Regression and LightGBM Classifier
- Evaluated models using precision, recall, F1-score, and AUC-ROC
- Selected LightGBM as the final model based on performance metrics

Real-Time Prediction:

- The deployed model allows real-time prediction of donation probability for each potential donor
- Enables targeted outreach to donors most likely to donate

Thank you

